

SOC ALERT SIMULATION REPORT

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1. Incident Overview

During continuous monitoring of system security logs, multiple failed login attempts were identified on a Windows system. These repeated failures suggest a potential unauthorized access attempt targeting the system environment.

2. Objective

The primary objective was to simulate suspicious login activity and analyze Windows Security Logs to understand how Security Operations Center (SOC) alerts are triggered, monitored, and investigated in real-world scenarios.

3. Methodology

A controlled security simulation was conducted by intentionally generating multiple failed login attempts on a Windows machine. The generated telemetry was monitored using the Windows Event Viewer, specifically focusing on the Security logs path: *Event Viewer* → *Windows Logs* → *Security*.

4. Observations

- **Detection Count:** A total of 6 failed login attempts were captured.
- **Core Event ID:** Event ID 4625 (Failed Logon).
- **Classification:** Categorized as Audit Failure.
- **Timeline:** All attempts occurred within a notably short time interval.
- **Target System Account:** SYSTEM / MSI\$.
- **Technical Status Codes:** 0xC000006D (Bad credentials) and 0xC0000380 (Authentication failure detail).

5. Identified Issue: Brute Force Attack

The pattern of repeated, rapid failed login attempts is a classic indicator of a **Brute Force Attack Pattern**. In this scenario, an actor attempts multiple password combinations in an effort to gain unauthorized access to the system.

6. Affected System

System Architecture: Windows Machine

Computer Name: MSI

Log Source: Microsoft Windows Security Logs

7. Evidence Collection

- **Event ID:** 4625 [Audit Failure]
- **Quantity:** 6 sequential attempts
- **Log Messaging:** "An account failed to log on"
- **Source:** Microsoft Windows Security Auditing

8. Impact & Risk Analysis

- Elevated risk of unauthorized system access and privilege escalation.
- Potential for total account compromise if the attack is not mitigated.
- Vulnerability to automated password-based attacks.
- Indicates a need for improved authentication security controls.

9. Response & Mitigation Strategy

The following defensive actions are recommended to mitigate such threats:

- **Account Lockout:** Enable policies that lock accounts after a specified number of failed attempts.
- **Password Complexity:** Enforce strong, non-repeating password policies.
- **MFA:** Implement Multi-Factor Authentication (MFA) to provide an additional layer of security.
- **SIEM Integration:** Use SIEM tools to monitor logs for real-time alerting and automated response.

10. Conclusion

The simulation successfully illustrated a brute force attack scenario using Event ID 4625. This project demonstrates the critical role of SOC analysts in identifying suspicious authentication behavior and implementing proactive security measures to prevent breaches.

Mini-Project Report - Prepared by P. Kaalai Selvam